INCH-POUND

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31 March 1989
SUPERSEDING
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7 January 1977

MILITARY SPECIFICATION

ANTENNA, AS-1735( )/SRC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the ultra high frequency (UHF) Antenna, AS-1735( )/SRC, hereinafter referred to as the antenna, a vertically polarized, four element array, providing omnidirectional azimuth coverage over a frequency range from 225 megahertz (MHz) to 400 MHz when mounted around a central supporting mast ranging from 101.6 millimeters (mm) (4 inches (in.)) to 381.1 mm (15 in.) in diameter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space And Naval Warfare Systems Command, SPAWAR PMW 152, Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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SPECIFICATIONS

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-2036	Enclosures For Electric And Electronic Equipment
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-B-007883	Brazing Of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum And Aluminum Alloys
MIL-W-8604	Welding, Fusion; Aluminum Alloys; Process And Performance Of
MIL-P-15024/5	Plates, Identification
MIL-E-16400G	Electronic, Interior Communications And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, Provisioned Items (Repair Parts): Packaging Of

STANDARDS

MILITARY

MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-781	Reliability Testing For Engineering Development, Qualification And Production
MIL-STD-810C	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1310	Shipboard Bonding, Grounding, And Other Techniques For Electromagnetic Compatibility And Safety

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MIL-STD-2000	Standard Requirements For Soldered Electrical And Electronic Assemblies
MS 91610	Connector, Receptacle, Electrical, Series LC, UG-352/U

HANDBOOK

MILITARY

MIL-HDBK-781	Reliability Test Methods, Plans, And Environments For Enginnering Development, Qualification, And Production

2.1.2 Other Government drawings and publications. The following other Government drawing and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

NAVELEX 28687-0109350	AS-1735/SRC VHF Antenna Fabrication Plan

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

DDS 072-1	Shock Design Values
TE000-AB-GTP-010	Parts Application, And Reliability Information Manual For Navy Electronic Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C-95-2-82 Radio Frequency Radiation Hazard Warning Symbol

(Application for copies should be addressed to American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The antenna shall be in accordance with MIL-E-16400G to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.1.2 Government-furnished material (GFM). GFM, if any, including either Government-furnished information (GFI) or Government-furnished equipment (GFE), shall be as specified in the contract.

3.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with the Parts, materials, and processes paragraph of MIL-E-16400G, to the extent specified herein.

3.2.1 Parts. Selection of parts shall be in accordance with the Selection paragraph of MIL-E-16400G, unless otherwise specified herein.

3.2.1.1 Parts control. The parts to be incorporated in the antenna shall be selected in accordance with MIL-STD-965, Procedure I, except as otherwise specified herein. SPAWAR Drawing 28687-0109350 shall be used for guidance only.

3.2.1.2 Interchangeability. Unless otherwise specified in the contract, antennas shall be interchangeable as specified in MIL-STD-454, Requirement 7, with corresponding items of the approved product baseline.

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3.2.1.3 Derating. Temperature, power, voltage stress levels and ratios, as applicable, as well as physical stresses, strengths, and processes shall be explicitly included in all mean-time-between-failures (MTBF) estimates. The antenna design shall adhere to the derating guidelines specified in NAVSEA TE000-AB-GTP-010.

3.2.1.4 Connectors, radio frequency (RF). The RF input to the antenna shall be by a connector capable of mating with a Series LC connector (Type UG-352/U or comparable) in accordance with MS 91610.

3.2.2 Materials. Materials shall be as specified in 3.2.2.1 and 3.2.2.2.

3.2.2.1 Selection of materials. Where metal-to-metal contact of dissimilar metals occurs, the selection of materials shall be in accordance with the Dissimilar metals paragraph of MIL-E-16400G, for combinations of metals subject to immersion, splashing, or spray from sea water.

3.2.2.2 Unacceptable materials. The antenna design shall comply with the Unacceptable materials paragraph of MIL-E-16400G. The antenna design shall not include the materials specified in a through g:

a. Flammable or combustible materials
b. Polychlorinated biphenyls
c. Asbestos and asbestos compounds
d. Fragile or brittle materials
e. Beryllium and beryllium compounds unless so identified
f. Lithium and lithium compounds unless specifically approved by the procuring activity
g. Chopped fiber

3.2.3 Processes. Processes shall be as specified in 3.2.3.1 through 3.2.3.7.

3.2.3.1 Painting. The paint system used for the antenna shall conform to the Adhesion and blister resistance paragraph of MIL-E-16400G.

3.2.3.2 Exposed equipment (exterior use). The antenna system shall be in accordance with the Painting paragraph of MIL-E-16400G for exposed equipment (exterior use).

3.2.3.3 Castings. All castings shall be in accordance with the Castings paragraph of MIL-E-16400G.

3.2.3.4 Brazing. Brazing shall conform to MIL-B-007883, Grade A joint qualities.

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3.2.3.5 Corrosion protection and corrosion-resistant treatments. All aluminum structures and assemblies used in the antenna shall be protected from corrosion by a chemical conversion treatment in accordance with MIL-C-5541, Class 3.

3.2.3.6 Welds. Structural welding shall conform to MIL-W-8604, Class A.

3.2.3.7 Soldering. Soldering procedures, and soldering processes shall be in accordance with MIL-STD-2000.

3.3 Design. Antenna design shall be as specified in 3.3.1 through 3.3.5.3.

3.3.1 General design and construction. The UHF antenna shall be a single, interconnected unit consisting of four individual dipole elements mounted at 90-degree intervals in a horizontal plane about a central support mast. Internal interconnecting RF feed lines and impedance matching sections shall permit simultaneous in-phase excitation of all four dipole elements from a single RF input connector. The antenna shall be adaptable for installation around a central support mast ranging between 101.6 mm (4 in.) to 304.8 mm (11 in.) in diameter and shall have adapter assemblies capable of allowing installations on masts sizes of 304.8 mm (11 in.) to 381.0 mm (15 in.) in diameter. The input impedance shall be 50 ohms over the frequency range of 225 MHz to 400 MHz.

3.3.2 Reliability design. The design of the antenna shall minimize failures due to parameter change, including drift, as a function of manufacturing tolerances, environmental factors, and age. The factors specified in a through h shall be explicitly considered:

a. Normal manufacturing tolerances including soldering, wiring, placement, and machinery tolerances
b. Normal variations in material quality, including finish tolerances
c. Range of ambient temperatures and their rate of change
d. Vibration levels, steady-state and transient
e. Humidity
f. Operational aging, particularly that of dielectric material from exposure to sun, ozone, and stack gasses
g. Nonoperational aging (storage life), shipping, and handling effect
h. Salt spray and stack gas deposits, particularly those on insulators

3.3.3 Maintenance design. The antenna shall be constructed so that maintenance and repair can be made with a minimum of disturbance to other parts. The antenna shall be readily demountable and interchangeable to provide ease of servicing, maintenance, and replacement.

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3.3.4 Safety. Safety criteria shall be in accordance with the Safety criteria paragraph of MIL-E-16400G.

3.3.5 Mechanical design. The mechanical design of the antenna shall conform to the Rounded corners and edges and Moisture pockets paragraphs of MIL-E-16400G. Antiseize compounds or silicon lubricant shall be used on all bolted or screwed joints, flanges, O-rings, and gaskets as appropriate, to facilitate the maintenance and serviceability of the antenna.

3.3.5.1 Enclosure. The degree of enclosure for the antenna shall be watertight as specified in MIL-STD-108.

3.3.5.2 Weight and markings. Equipment weighing in excess of 15.88 kilograms (35 pounds) shall have the weight clearly marked on the external surface of the equipment and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied. Identification nameplates for the antenna shall be affixed in a conspicuous location. Nameplate format, material, lettering application, and information shall conform to MIL-P-15024/5.

3.3.5.3 Magnetic materials. The antenna shall have a minimum magnetic signature in accordance with the requirements specified in the Magnetic materials paragraph of MIL-E-16400G.

3.4 Performance requirements. The antenna shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to adverse conditions normally encountered in Naval shipboard communication service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance.

3.4.1 Radiation pattern. The maximum radiation pattern shall occur at zero elevation angle over the 225-MHz to 400-MHz frequency band. The horizontal pattern of this antenna will be omnidirectional to within ±2 decibels (dB) over the entire 225-MHz to 400-MHz frequency range when mounted around a central support mast 101.6 mm (4 in.) to 279.4 mm (11 in.) in diameter and within ±4 dB when mounted around a central support mast 381.00 mm (15 in.) in diameter.

3.4.2 Frequency range. The antenna and feed-system shall operate over the frequency range of 225 MHz to 400 MHz.

3.4.3 Polarization. The antenna shall transmit or receive vertically polarized RF energy.

3.4.4 Gain. The antenna gain shall be 2.1 ±0.25 dB above an isotropic level.

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3.4.5 Power rating (RF). The antenna shall be capable of handling 2000 watts peak-envelope RF power over the frequency range.

3.4.6 Voltage standing wave ratio (VSWR). The impedance at the input to the antenna shall match the 50-ohm impedance of the input coaxial cable so that the VSWR does not exceed 1.8:1 over the entire 225-MHz to 400-MHz range when mounted.

3.5 Environmental requirements. The antenna, exposed to weather (unsheltered), shall be capable of continuous reliable operation, within the performance limits specified herein, under the environmental requirements specified in 3.5.1 through 3.5.13.

3.5.1 Temperature, operating. The antenna shall be capable of conforming to all performance characteristics specified herein while operating at any temperature in the range -28° Celsius (C) (-18.4° Fahrenheit (F)) to +65°C (149.0°F) in accordance with Range 2 of MIL-E-16400G, for unsheltered equipment.

3.5.2 Temperature, nonoperating. The antenna shall be capable of conforming to all performance characteristics specified herein after long periods of nonoperation at ambient air temperatures in the range -62°C (-79.6°F) to +71°C (160.0°F) in accordance with Range 2 of MIL-E-16400G, for unsheltered equipment.

3.5.3 Humidity. The antenna shall have full operational capability when subjected to a relative humidity of 0 percent to 100 percent.

3.5.4 Salt fog (spray). The antenna shall be tested as specified in 4.6.6.10.

3.5.5 Wind velocity. The antenna shall operate normally in winds having a relative velocity of 185 kilometers per hour (km/hr) (100 knots) and shall withstand, without damage, winds having a relative velocity as great as 222 km/hr (120 knots).

3.5.6 Snow and ice loading. The antenna shall be designed to operate with a snow and ice loading of 271 kilograms per square meter (55.5 pounds per square foot). Performance includes signal propagation; due allowance will be granted for discrepancies as would normally be expected through snow and ice.

3.5.7 Pressurization. Normally the antenna shall be operated unpressurized, but shall be capable of being pressurized to 10 pounds per square in. (psi) and show no signs of leaking when immersed in water.

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3.5.8 Moisture leakage. The antenna feeder system assembly shall be of vapor tight construction to prevent antenna performance degradation due to ingress of moisture to the feeder system. Assembly of coaxial cables, connector fittings, RF transmissions, and parts shall be vapor sealed and watertight at ends and junctions as specified in MIL-STD-108 and MIL-E-2036. The antenna feeder system vapor seals shall be capable of conforming to the requirements of the leakage test specified in 4.6.6.2.

3.5.9 Fungus. The antenna shall be tested as specified in 4.6.6.8.

3.5.10 Shock. The antenna shall be capable of withstanding, in the vertical orientation, the Grade A, Class I, Type A shock test for mast-mounted equipment of MIL-S-901.

3.5.11 Vibration. The antenna shall be capable of withstanding the vibrations of MIL-STD-167-1, Type I. The upper frequency limit shall be performed at 50 hertz (Hz).

3.5.12 Nuclear air blast. The antenna shall conform to the Nuclear air blast paragraph of MIL-E-16400G. The peak overpressure, P, shall be 7.2 psi and the peak dynamic pressure, Q, shall be 1.2 psi.

3.5.13 Solar radiation. The antenna shall maintain the performance specified herein after being subjected to the solar radiation test of 4.6.6.12.

3.6 Electromagnetic interference suppression. The antenna shall have provisions for post-installation bonding and grounding in accordance with the requirements of MIL-STD-1310.

3.7 Transportability. All antenna equipment shall be transportable by air, rail, truck, barge, or ship.

3.8 Electric shock hazard labeling. Hazard labels shall be permanently attached to each antenna in a position readily visible to personnel in the area and shall contain this warning in the upper triangle:

DANGER

RADIO FREQUENCY SHOCK HAZARD

This statement shall appear in the lower triangle:

A RADIO FREQUENCY SHOCK HAZARD MAY EXIST
TO PERSONNEL CONTACTING THE ANTENNA

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Color and format of the labels shall conform to ANSI C-95-2-82 requirements. Dimension D as specified in the Radio frequency radiation hazard warning symbol figure of ANSI C-95-2-82, shall be at least 0.3195 centimeter (0.125 in.).

3.9 Reliability and maintainability. Reliability and maintainability shall be as specified in 3.9.1 and 3.9.2.

3.9.1 MTBF. The MTBF for the antenna shall be not less than 25,000 hours.

3.9.2 Mean-time-to-repair (MTTR). Under sea conditions not greater than Sea State 4 and excluding rerigging, the MTTR for corrective maintenance to the antenna shall be not greater than 2 hours (see 6.4.3).

3.9.3 Maximum repair time (M_{max}). Under sea conditions not greater than Sea State 4 and excluding rerigging, the M_{max} shall be not greater than 4 hours (see 6.4.2).

3.10 Workmanship. Workmanship shall be as specified in 3.10.1 and 3.10.2.

3.10.1 General workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.10.2 Random vibration. All antennas shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor from the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through d:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)
    1. In-process inspection (see 4.5.1)
    2. Production inspection (Group A) (see 4.5.2)
    3. Production control inspection (Group B) (see 4.5.3)
    4. Environmental inspection (Group C) (see 4.5.4)
c. Safety verification (see 4.6.7)
d. Inspection of packaging (see 4.9)

4.3 Standard test conditions. Unless otherwise specified herein, standard test conditions shall be as specified in MIL-E-16400G.

4.4 First article inspection. Unless otherwise specified (see 6.2), four units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

4.5 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.5.1 through 4.5.6.

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TABLE I. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process inspection	Group A	Group B	Group C
Interchangeability	3.2.1.2	4.6.1	X			X	
Parts, materials, and processes	3.2, 3.2.1, 3.2.2, 3.2.3	4.6.1	X				
Selection of parts	3.2.1	4.6.1	X				
Parts control	3.2.1.1	4.6.1	X				
Connectors, RF	3.2.1.4	4.6.1	X	X			
Selection of materials	3.2.2.1	4.6.1	X				
Unacceptable materials	3.2.2.2	4.6.1	X				
Antenna design	3.3, 3.3.3	4.6	X				
Safety	3.3.4	4.6.7	X	X			
Dimensions and weight	3.3.1, 3.3.5.2	4.6.1	X		X		
Identification plates	3.3.5.2	4.6.1	X		X		
Information plates	3.9.3	4.6.1	X	X			
Workmanship	3.10	4.6.1	X		X		
Random vibration	3.10.2	4.8.2	X			X	X
Frequency range	3.4.2	4.6.4	X		X		X
Impedance	3.4.6	4.6.4	X		X		X
Power rating (RF)	3.4.5	4.6.4	X				X
VSWR	3.4.6	4.6.4	X		X		X
Horizontal pattern	3.4.1	4.6.4	X				X
Polarization	3.4.3	4.6.4	X				X

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TABLE I. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection			
				In-process inspection	Group A	Group B	Group C
Gain	3.4.4	4.6.4	X		X		X
Wind velocity	3.5.5	4.6.6.9	X				
Snow and ice loading	3.5.6	4.6.6.9	X				
Magnetic materials	3.3.5.3	4.6.3.3	X				
Welds	3.2.3.6	4.6.3.1	X	X			
Brazing	3.2.3.4	4.6.3.2	X	X		X	
Enclosure	3.3.5.1	4.6.1	X			X	
Environmental requirements	3.5	4.6.6	X			X	
Temperature, operating	3.5.1	4.6.6.4	X			X	
Temperature, nonoperating	3.5.2	4.6.6.3	X			X	
Humidity	3.5.3	4.6.6.7	X			X	
Salt fog (spray)	3.5.4	4.6.6.10	X			X	
Leakage (immersion)	3.5.7, 3.5.8	4.6.6.2	X			X	
Shock	3.5.10	4.6.6.5	X				
Vibration	3.5.11	4.6.6.6	X				
Nuclear air blast	3.5.12	4.6.6.11	X				
Solar radiation	3.5.13	4.6.6.12	X				
Fungus	3.5.9	4.6.6.8	X				
Adhesion and blister resistance	3.2.3.1	4.6.2	X				
Reliability	3.9.1	4.7	X				

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4.5.1 In-process inspection. In-process inspection shall be performed on every antenna offered for delivery. In-process inspection shall be performed during, or where a process is performed on identical or representative samples, or prior to the final manufacture of an item. In-process inspection shall be comprised of examinations, tests, and demonstrations to prove the quality of workmanship and to verify conformance to specified requirements, for example, tests and demonstrations to detect departure from prescribed processes. In-process inspection shall include the examinations and tests specified in the in-process inspection column of TABLE I.

4.5.2 Production inspection (Group A). Production inspection shall be performed on every antenna offered for delivery. Production inspection shall comprise examinations and tests to prove the quality of workmanship and to reveal omissions and errors of the production process, for example, tests to detect deviations from design, and tests to detect hidden defects in materials. Production inspection shall include the examinations and tests specified in Group A of TABLE I. Random vibration shall be the first inspection performed.

4.5.3 Production control inspection (Group B). Production control inspection shall be performed on antennas that have passed production inspection specified in 4.5.2. Lot acceptance criteria shall be based on a single sampling plan with a zero acceptance number. Production control inspection shall include the examinations and tests specified in Group B of TABLE I. The antenna shall satisfactorily conform to the requirements of production control inspection prior to releasing the lot for shipment.

4.5.3.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.4 Environmental inspection (Group C). Environmental inspection shall be performed on a sampling basis. Environmental inspection shall encompass environmental tests to prove the durability of the antenna. Environmental inspection shall include the examinations and tests specified in Group C of TABLE I. Unless otherwise specified (see 6.2), the basis for sampling shall be as specified in a and b:

a. When a first article test is performed, no additional testing in Group C is required.

b. When first article inspection is waived, Group C testing shall be performed on one antenna that has completed, and has passed, the Group B inspection specified in 4.5.3.

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4.5.5 Nonconforming production control and environmental sample units. If a sample unit fails the inspections specified in 4.5.3 and 4.5.4, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative (QAR) the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to the specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.5.6 Reinspection of conforming production control and environmental inspection samples. Unless otherwise specified (see 6.2), sample units which have been subjected to, and have passed, production control or environmental inspection, or both, may be accepted on the contract provided they are resubjected to, and pass, production inspection for workmanship after repair of all defects and damage incident to the Group B or Group C inspections.

4.6 Test methods. Test methods shall be in accordance with MIL-E-16400G, except as specified in 4.6.1 through 4.6.6.12.

4.6.1 Surface examinations. The antenna shall be examined to the extent necessary to determine conformance to the applicable requirements of this specification. The antenna shall be examined to determine conformance to the requirements specified in a through i:

a. Parts, materials, and processes (see 3.2)
b. Finish (see 3.2.3.2)
c. Brazing (see 3.2.3.4)
d. Welding (see 3.2.3.6)
e. Dimensions and weight (see 3.3.1 and 3.3.5.2)
f. Safety (see 3.3.4)
g. Enclosures (see 3.3.5.1)
h. Marking (see 3.3.5.2)
i. Workmanship (see 3.1

In the process of examination, the antenna shall not be disassembled in a manner that will affect the performance, durability, or appearance of the antenna.

4.6.2 Adhesion and blister resistance. The paint system test shall be performed in accordance with the Paint system test paragraph of MIL-E-16400G, Procedure II.

4.6.3 Mechanical and structural tests. Mechanical and structural tests shall be as specified in 4.6.3.1 through 4.6.3.3.

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4.6.3.1 Weld test. The process of fabricating welds employed in the construction of the antenna components shall be tested to determine conformance to the requirements of 3.2.3.6. Testing shall be accomplished prior to fabrication of the first article or production antenna.

4.6.3.2 Brazing test. The brazing test shall be performed to determine conformance to the requirements of 3.2.3.4.

4.6.3.3 Magnetic materials test. The antenna shall be tested in accordance with the Magnetic material test paragraph of MIL-E-16400G to determine conformance to 3.3.5.3.

4.6.4 Electrical tests. Electrical tests shall be taken across the 225-MHz to 400-MHz band using sweep frequency instrumentations in accordance with contractor test procedures approved by the procuring activity. The sequence of the tests shall be as specified in a through f:

a. Impedance
b. Power handling capability
c. VSWR
d. Horizontal and vertical patterns
e. Polarization
f. Gain

4.6.5 Test sequence. The antenna shall be subjected to the performance tests specified in 4.6.4 and 4.6.6. Electrical tests shall be performed prior to, and following, the completion of environmental testing.

4.6.6 Environmental tests. Environmental tests shall be as specified in 4.6.6.1 through 4.6.6.12.

4.6.6.1 Environmental test sequence. Unless otherwise specified herein, environmental tests shall be performed in the sequence specified in a through l:

a. Leakage (immersion) test in accordance with 4.6.6.2
b. Low temperature test in accordance with 4.6.6.3
c. High temperature test in accordance with 4.6.6.4
d. Solar radiation (sunshine) test in accordance with 4.6.6.12
e. Leakage (immersion) test in accordance with 4.6.6.2
f. Humidity test in accordance with 4.6.6.7
g. Fungus test in accordance with 4.6.6.8
h. Salt fog (spray) test in accordance with 4.6.6.10
i. Shock test in accordance with 4.6.6.5

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j. Vibration test in accordance with 4.6.6.6
k. Nuclear air blast in accordance with 4.6.6.11
l. Leakage (immersion) test in accordance with 4.6.6.2

The leakage (immersion) test is repeated in order to ensure that the watertight integrity of the antenna has not been impaired during selected environmental tests.

4.6.6.2 Leakage (immersion) test. The leakage (immersion) test shall be performed as specified herein. The antenna shall be pressurized to 10 psi and immersed into a tank of water. There shall be no bubbles visible in the water.

4.6.6.3 Low temperature. Conformance to the temperature requirements for operating and nonoperating conditions shall be verified in accordance with the Temperature test sequence and Low temperature paragraphs of MIL-E-16400G.

4.6.6.4 High temperature. Conformance to the temperature requirements for operating and nonoperating conditions shall be verified in accordance with the Temperature test sequence and High temperature paragraphs of MIL-E-16400G.

4.6.6.5 Shock test. The antenna shall be tested to determine conformance to the requirements of 3.5.10 using the floating shock platform modified with a yardarm test fixture. The antenna shall be mounted in a configuration such that when two antennas are mounted, the bottom edge of the lower dipole is at least 2.032 centimeters (cm) (8 in.) higher than the yardarm and there is a 3.48-cm (12-in.) space between the upper edge of the lower dipole and the lower edge of the upper dipole. The upper edge of the upper dipole should be at least flush with the mounting pole though the mounting pole may exceed the height. The antenna shall be functioning after the test.

4.6.6.6 Vibration test. The vibration test shall be in accordance with the Vibration test paragraph of MIL-E-16400G, and the Variable frequency test and the Endurance test for mast mounted equipment paragraphs of MIL-STD-167-1.

4.6.6.7 Humidity test. Conformance to the humidity requirements of 3.5.3 shall be verified in accordance with the Humidity test paragraph of MIL-E-16400G.

4.6.6.8 Fungus test. The fungus test shall be performed in accordance with MIL-STD-810C, Method 508.2.

4.6.6.9 Environmental analysis. Conformance to the requirements of 3.5.5 and 3.5.6 shall be verified by analysis to determine that the antenna is capable of sustaining the specified forces while conforming to all the performance requirements.

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4.6.6.10 Salt fog test. Tests shall be performed in accordance with the Salt fog test paragraph of MIL-E-16400G, for exposed equipment, using Procedure II. The test shall verify conformance to the painting processes of 3.2.3.1. The criteria of the Adhesion and blister resistance paragraph of MIL-E-16400G shall be used for evaluation of the painting processes.

4.6.6.11 Nuclear air blast calculations. The antenna shall be tested to determine conformance to the requirements of 3.5.12.

4.6.6.12 Solar radiation (sunshine). The solar radiation (sunshine) test shall be performed in accordance with MIL-STD-810C, Method 505.1, Procedure II.

4.6.7 Safety verification. Examinations, analyses, and tests shall be performed to verify conformance to 3.3.4.

4.7 Reliability and maintainability. Examinations, analyses, and tests shall be performed to verify conformance to 3.9.

4.7.1 Reliability. Conformance to the requirements of 3.9.1 shall be determined by a fixed duration test using a minimum of four units at a period of 1000 hours, each unit in a combined test environment as specified in the Externally mounted equipment paragraph of MIL-HDBK-781. The requirements of Task 301, and the general requirements of MIL-STD-781 shall apply. One chargeable failure shall result in a reject decision and require the performance of corrective action.

4.7.2 Maintainability. Conformance to the requirements of 3.9.2 shall be demonstrated by analysis.

4.8 Workmanship. Workmanship shall be as specified in 4.8.1 and 4.8.2.

4.8.1 General workmanship. The antenna shall be examined for conformance to the requirements of 3.10.1.

4.8.2 Random vibration. Each antenna shall be subjected to vibration in turn along each of three rectilinear orientation axes of the equipment; vertical, side-to-side, and front-to-back. Two or three axes simultaneous vibration may also be used. A vibration survey shall be performed at an amplitude lower than that shown on FIGURE 1 to determine that no damaging levels of response are present. The antenna may require amplitude notching to successfully perform the vibration screen at the levels shown on FIGURE 1. The procuring activity shall be advised of any required changes to FIGURE 1. The vibration shall be performed at the end item level. All the hardware, including clamps and connectors, shall be exposed to vibration to determine conformance to the requirements of 3.10.2. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in each axis.

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All items shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 1. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analyses of the acceleration spectral density (ASD). The instantaneous acceleration peaks may be limited to three times the root-mean-square acceleration level. The antenna shall be pressurized to 10 psi prior to starting the vibration. Pressure shall be monitored throughout the vibration. Any detectable changes in pressure shall be considered a failure. VSWR shall be monitored during the vibration as a means of assessing the mechanical and electrical integrity of the antenna. All failures occurring during vibration shall be corrected and the vibration continued for a cummulative time of 10 minutes in each axis.

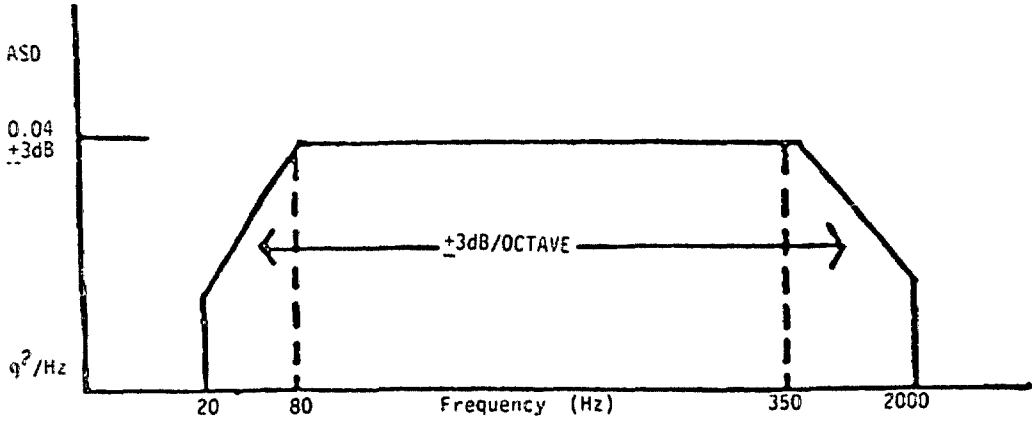


FIGURE 1. Random vibration curve.

4.9 Inspection of packaging. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna, covered by this specification, is intended for use on board ship and at shore base installations to provide vertically polarized horizontal omnidirectional coverage over a frequency range from 225 MHz to 400 MHz.

6.2 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. GFM, if any (see 3.1.2)
c. Interchangeability requirements if other than specified (see 3.2.1.2)
d. Number of first article samples to be submitted if other than specified in 4.4
e. If sampling and equipment for Group C inspection is other than specified in 4.5.4
f. When reinspected conforming production control and environmental inspection samples may not be accepted (see 4.5.6)
g. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.3.

6.4.1 Shock, Grade A. Shock, Grade A, items are defined as machinery, equipment, and systems essential for the safety and continued combat capability of the ship. Shock testing is to demonstrate that the item will continue to perform its principal function without significant change in performance and that no portion of the equipment will come adrift or otherwise become a hazard to personnel or to Grade A machinery, equipment, or systems during, and following, the application of shock loadings.

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6.4.2 M_{max}. M_{max} is the maximum-time-to-repair at the 95th percentile. That is, 95 percent of all maintenance tasks should be accomplished within this time (see 3.9.3).

6.4.3 MTTR. MTTR is the summation of active repair time during a given period of time divided by the total number of malfunctions during the same interval. Active repair time includes the following task time increments: preparation, localization, isolation, disassembly, interchange, reassembly, alignment, and verification checkout (see 3.9.2).

6.5 Production control and environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.5).

6.6 Personnel skills. In designing equipment for ease of maintenance, information concerning Naval personnel skills and skill levels is available in NAVPERS 18068 and NAVPERS 15105. Electronics Technician or Radiomen ratings to be considered in performing maintenance tasks are Petty Officer Third Class (E-4) and Seaman Striker (E-3) (Class A school graduate).

6.7 GFM, GFI, and GFE. The contracting officer should arrange to furnish the property specified in 3.1.2.

6.8 Subject term (key word) listing.

Antenna, AS-1735( )/SRC
AS-1735( )/SRC, Antenna
Intended use: Shipboard and shore-based

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
NAVY - EC

(Project No. 5985-N593(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-24233B(EC)	Antenna, AS-1735( )/SRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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